INTERIM FEDERAL SPECIFICATION

CLAMPING TOOL, STRAP BAND, HOSE

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope.- This specification covers one type of clamping tool used in applying metal strapping to pliable tubing, when fixing it to rigid tubing, etc. (see 6.1, intended use).

1.1.1 Federal specification coverage.- Federal specifications do not include all varieties of the commodity indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.- The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:
Fed. Std. No. 102 - Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:
MIL-H-15424 - Hand Tools; Packaging of.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustration.- Except for dimensional requirements, the illustration shown herein is for the convenience of identification and is not intended to preclude the purchase of clamping tools which are otherwise in accordance with the requirements of this specification.

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3.2 Material.- Clamping tools shall be made of materials of sufficient strength and hardness to meet the tests specified in section 4. The materials shall be free of defects, which would be detrimental to the use or serviceability of the clamping tool.

3.3 Construction.- The clamping tools shall be constructed to withstand the stresses of normal application, without fracture or deformation.

3.4 Design.- The clamping tool shall be similar to figure 1. The tool shall be capable of providing tension forces, from 0 to 3,000 lbs, to steel strapping. Component parts of the tool shall be interchangeable.

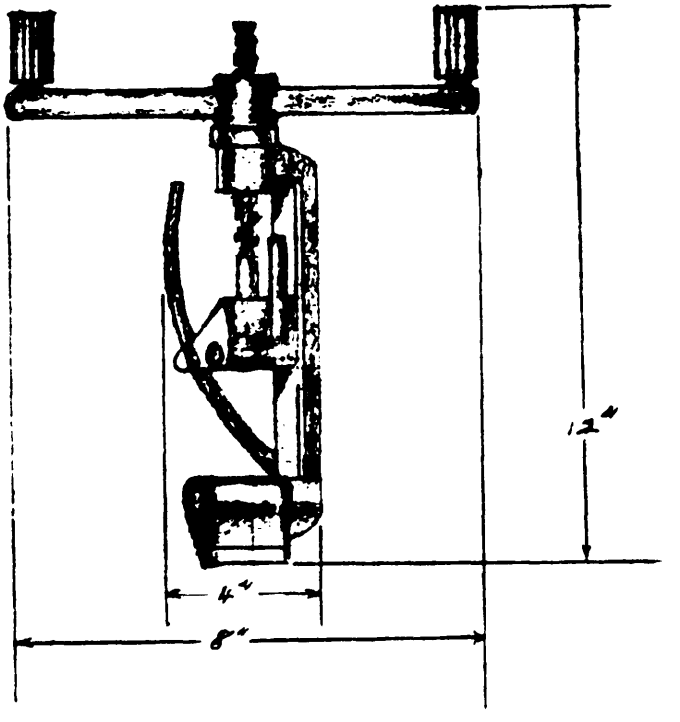


Figure 1. - Clamping Tool, Strap Band, Hose

3.4.1 Performance.- The clamping tool shall be capable of tightening and cutting steel strapping placed around a two inch rubber hose, assembled over a rigid male fitting, in the maximum time of one minute.

3.4.2 Dimensions.- The maximum dimensions shall be as shown in figure 1. The strapping clamp head shall travel three and one-half inches plus or minus 1/4 inch.

3.4.3 Distortion resistance.- The clamping tool shall withstand 3,000 lbs. tension, for five minutes, without fracture or distortion. There shall be no slippage of the strap gripped by the jaws during this test (see 4.5.1).

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3.4.4 Strap cutting.- The clamping tool shall be capable of easily cutting steel strapping, at any tension of from 1 to 3,000 lbs., at a distance of not less than 1 inch nor more than 1-1/4 inch from the strapping clamp or buckle.

3.4.5 Durability.- The clamping tool shall provide a minimum of 2,000 hours of use without repair, providing it is properly used and maintained (see 4.5.2).

3.4.6 Weight.- The maximum weight of the clamping tool shall be five pounds.

3.4.7 Strap sizes.- The strapping tool shall be suitable for tensioning metal strapping of the following range of sizes:

3/8 thru 3/4 inch in width and 0.025 thru 0.030 inch in thickness.

3.4.8 Hardness.- The Rockwell hardness of the cutter and jaws shall be as follows:

Stationary cutter	- "C"	60 - 65
Movable cutter	- "C"	60 - 65
Stationary jaw	- "B"	90 - 95
Movable serrated jaw	- "C"	63 - 65

3.5 Finish.- The surfaces shall be free from cracks, blisters, pits, scale or foreign substance.

3.5.1 Coating.- The entire visible surface shall be electroplated with zinc, nickel, or cadmium.

3.6 Marking.- The clamping tool shall be marked in a plain and permanent manner with the manufacturers name, symbol, or trademark of such known character that the source of manufacture may be readily determined.

3.7 Workmanship.- The workmanship shall be first class throughout.

3.8 Instructions for operation and maintenance.- Complete instructions on the method of operation and maintenance of the clamping tool shall be furnished with each unit.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components.- In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or if none, in accordance with this specification.

4.2 Sampling for inspection and acceptance.- Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in Military Standard MIL-STD-105, except where otherwise indicated herein.

4.2.1 Sample unit and lot formation.- The sample unit shall be one clamping tool for the purpose of lot formation. A lot shall be comprised of all of the clamping tools submitted for inspection at one time, which were manufactured by one source.

4.2.2 Sampling for visual examination.- Each sample unit shall be visually examined in accordance with 4.4. The inspection level shall be level II with an AQL of 4.0 defects per hundred units.

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4.2.3 Sampling for dimensional and weight examination.- Each sample unit shall be examined in accordance with 4.5. The inspection level shall be level S-3 with an AQL of 2.5 defects per hundred units.

4.2.4 Sampling for tests.- Each sample unit shall be tested in accordance with 4.5. The inspection level shall be level S-3 with an AQL of 1.5 defects per hundred units.

4.3 Visual examination.- Each sample unit shall be examined for visual deviations from specified requirements in material, construction, design, finish, marking and workmanship. Any deviation from these requirements is a defect.

4.4 Dimensional and weight examination.- Each sample unit shall be examined for conformance with dimensional and weight requirements. Any deviation from requirements is a defect.

4.5 Testing.- Each sample unit selected in accordance with 4.2.4 shall be subjected to tests in 4.5.1, 4.5.2 and 4.5.3.

4.5.1 Strength test.- The frame of the clamping tool shall be securely fastened in a stationary fixture. One end of a piece of strapping shall be placed in the jaws of the tool and the other end attached to a tension measuring device. A load of 3,000 pounds shall be developed by actuating the handle of the tension screw. The tool shall sustain this load for at least 5 minutes without deformation of any component or slippage of the strapping in the jaws. Each deformed component or slippage of strapping in the jaws is a defect.

4.5.2 Durability test.- Each sample tool shall be stressed as stated in 4.5.1, except that the load need not be maintained for five minutes and the strapping shall be severed by the cutting device when the maximum load is reached. This operation shall be repeated 100 times without the necessity for repair or replacement of any component of the tool. Each component that fails to comply with the above is a defect.

4.5.3 Hardness test.- The jaws and cutters of each sample selected for test shall be tested for conformance with hardness requirements in 3.4.8, using test method 243.1 of Federal Test Method Standard No. 151.

4.6 Inspection of preparation for delivery.- Preservation, packaging, packing and marking shall be inspected in accordance with requirements in section 4 of MIL-H-15424 for the level specified (see 6.2).

5. PREPARATION FOR DELIVERY
(For civil agencies, Fed. Std. No. 102 should be referred to for definitions and applications of the various levels of protection.)

5.1 Preservation and packaging.

5.1.1 Level A.- Preservation and packaging of clamping tools shall be in accordance with level A of MIL-H-15424 for puller, gear, and bushing, 2-1/4 thru 5-1/4 pounds.

5.1.2 Level C.- Preservation and packaging which will afford adequate protection against corrosion, deterioration, and damage during shipment from supply source to the first receiving activity for immediate use. (vendor to user)

5.2 Packing.

5.2.1 Level A.- Clamping tools, preserved and packaged as specified (see 6.2), shall be packed in accordance with level A requirements of MIL-H-15424 for puller, gear, and bushing, 2-1/4 thru 5-1/4 pounds.

5.2.2 Level C.- Packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

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5.3 Marking.

5.3.1 Civil agencies.- In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 Military agencies.- Marking shall be in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use.- The clamping tool covered by this specification is generally used for placing straps on pliable tubing when attaching it to rigid tubing; however, the tool is also used in splicing electric welding cable, tying conduit or cable to pipe lines or girders, and making cable line taps.

6.2 Ordering data.- Purchasers should exercise any desired option offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation, packaging and packing required (see 5.1 and 5.2).
(c) Marking, if different, (see 5.3).

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